Designing and Evaluating AI-Assisted Deductive Analysis Workflow for Education Impact Studies

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Introduction

- Public trust in research depends on the **methods**, not just the findings.
- LLMs increasingly assist qualitative coding, but their outputs are **probabilistic and often opaque**.
- Little evidence exists on how they perform **inside a real evaluation workflow**.

Purpose & Research Questions

Can AI be responsibly integrated into deductive qualitative coding while preserving **transparency**, **interpretive fidelity**, and **trust**?

1 Workflow design & evaluation
Can AI produce transparent, reviewable outputs that align with human judgment, and does iterative refinement improve reliability?

2 Model & configuration effects
How do model choice, prompt structure, and reasoning settings shape performance, transparency, and reproducibility?

3 Collaborative processes
How do researchers and data scientists turn expert judgment into structured guidance the AI can follow?

Setting

- Embedded in an ongoing, **multi-year mixed-methods evaluation** of organizational capacity to support institutional outcomes in postsecondary settings.
- **Human-labeled codebooks and summaries** are the reference, compared to AI outputs at every stage.

71	10	2021
INTERVIEWS	ORGS	STUDY YEAR

A 3-level deductive codebook

LEVEL	NAME	COUNT
GRANDPARENT	Role	4
PARENT	Capability	12
CHILD	Element	50

Example Element · Staffing & human capital
DEFINITION the org's own capacity to staff and run its programs.
WHAT TO LOOK FOR understaffing, unfilled roles, too few staff.

Deductive Analysis Workflow

A transparent pipeline that **mirrors how a human does deductive coding**.

1. **Interview transcripts** — with & without a set protocol →
2. **Q→A chunks** — split on speaker change + interviewer question →
3. **Coded evidence** — top-5 matches + the asked question as context feed the AI judge →
4. **AI codebook + findings** — findings, verbatim quotes + summaries →
5. **Researcher review** — validate, correct, finalize

A CLOSER LOOK: A MISCODE TRACED TO A GUIDANCE GAP · sharpening a "what to look for" (illustrative)

Q→A CHUNK
HOW: split on **speaker change**; each interviewer question paired with its answer

Interviewer: What's holding back your advising program?

Program Director: "We can't keep up with demand; students wait weeks for an appointment and some give up. We're down to two advisors for the whole campus."

→

CODED EVIDENCE
HOW: excerpt embedded → cosine **top-5** codebook shortlist → **AI-as-a-judge** picks the **best-fit** element, then applies a **relevance score** (1–10).

TOP-5 SHORTLIST FED TO THE JUDGE

▸ Service delivery capacity — AI pick · 7/10	
Staffing & human capital	Financial resources
Program management	Org. infrastructure

Judge's reasoning: "Excerpt centers on unmet demand and long appointment waits."

→

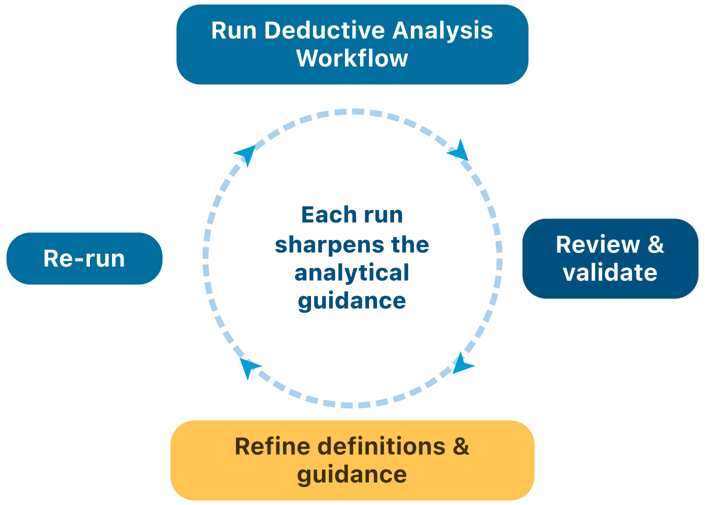
RESEARCHER REVIEW
HOW: reviewer catches the miscode, traces it to a gap in the element's guidance, and updates the codebook for the next run

OVERRIDE reassigned to **Staffing & human capital**

*Why: it overlaps with **Service delivery capacity**; Staffing's **"what to look for"** never drew the line between them.*

Added to **"what to look for"**: **"a service shortfall traced to too few staff." The re-run then codes it correctly.**

Human Oversight & Iteration



Later runs diverged most on **closely-related codes***; sharpening the analytical guidance led to the largest improvement.*

Researcher Review & Validation

AI outputs are **reviewable drafts, not final products**.

WHAT THEY CHECK

- Each **element assignment** and its stated reasoning
- **Dropped low-relevance excerpts and background text**, to confirm nothing important was excluded
- Whether findings faithfully reflect the overall interview

HOW IT FEEDS THE PROCESS

- Overrides on **closely-related codes** and borderline excerpts update the **analytical guidance** for the next run, so the workflow improves with use
- **Researcher effort** is heaviest during early calibration, then eases as outputs stabilize
- Researchers own the interpretation; data scientists own the technical implementation and guardrails behind it

Findings

WORKFLOW DESIGN & EVALUATION · RQ1
Structure and evaluation earned trust

1. **AI coding agreed with human coders:** element-level agreement on deliverable text reached 66% and word-level Krippendorff's α reached 0.71, and improved through iteration.
2. **Decomposition made each stage reviewable:** intermediate stages give natural human-review points that cut compounding errors, and let researchers trace findings to evidence and separate fixable errors from inherent ambiguity.
3. **Evaluation was continuous, not final:** evaluating alongside design surfaced failure modes early and drove targeted, stage-level fixes.

MODEL & CONFIGURATION · RQ2
Settings shaped behavior, not accuracy

1. **Models differed in behavior, not accuracy:** GPT-4.1, o3, and GPT-5 had similar accuracy measures, but diverged in formatting, verbosity, cost, and runtime.
2. **Deeper reasoning added detail, not accuracy:** it explained its picks more thoroughly but reached the same codes, and at times over-extrapolated.

COLLABORATION · RQ3
Co-design moved humans to judgment

1. **Designing outputs to fit researchers was pivotal:** shaping AI outputs around researchers' expertise and comfort (familiar, NVivo-style) greatly improved and sped the review process.
2. **Errors were guidance gaps,** not model failures ("is the model wrong, or does the guidance need refinement?").

Future Work

- **Testing across new contexts:** validate the workflow on new project types and codebook structures to map where the approach generalizes.
- **Multi-label coding:** let an excerpt carry more than one code, backed by retrieval that moves past the top-5 cosine shortlist to hierarchy-aware reranking.
- **Lighter researcher review:** explore ways to cut review and validation time, triaging excerpts by model confidence so the riskiest calls surface first with competing codes and reasoning inline.

Literature Cited

SCAN QR CODE
for the full reference list

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